SWIFTMART ECOMMERCE

Sales, Customer Performance & Churn Analysis

Professional Business Analytics Report

Portfolio Project

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EXECUTIVE SUMMARY

This comprehensive business analytics report presents a detailed analysis of SwiftMart, a leading Indian e-commerce platform specializing in electronics, books, toys, fashion, and consumer goods. The analysis spans transaction data from January 2022 through December 2025, encompassing 5,000 unique customers across 29,984 orders.

Metric	Value	Benchmark	Status
Total Revenue	₹939.8M	High	✓
Total Profit	₹253.5M	Strong (27% margin)	✓
Total Orders	29,984	Healthy Volume	✓
Total Customers	5,000	Established Base	✓
Churn Rate	12.04%	Critical (Target: <9%)	⚠️
Retention Rate	87.96%	Good	✓
Repeat Customers	83.88%	Excellent	✓
Avg Order Value	₹31,344	Growing	✓
Avg CLV	₹187,963	Strong	✓

The analysis reveals a mature, profitable e-commerce operation with strong revenue generation (₹939.8M) and solid profitability (₹253.5M, 27% margin). However, the 12.04% churn rate presents a critical opportunity for customer retention improvement. Key findings indicate that 83.88% of customers are repeat buyers, demonstrating strong product-market fit and customer loyalty among active buyers. The study identifies customer segmentation patterns, churn risk factors, and actionable recommendations for business growth.

PROJECT OVERVIEW

Project Scope

This portfolio project demonstrates advanced data analytics capabilities using Microsoft Excel, Power Query, Power Pivot, and dashboard visualization. The project transforms raw transactional data into actionable business intelligence through data integration, dimensional modeling, feature engineering, KPI development, and strategic recommendation.

Data Sources

Fact_Orders (29,984 records): Transaction records with sales, profit, and order details. Dim_Customers (5,000 records): Customer demographics and segmentation. Dim_Products (300 records): Product pricing and category information. Dim_Supportive_Cal (5,000 records): Calculated customer features including status, CLV, and churn indicators.

BUSINESS PROBLEM STATEMENT

While SwiftMart demonstrates strong financial performance, the organization faces four critical challenges: (1) Customer Churn Crisis—602 customers (12.04% churn rate) have become inactive, with 1,589 at-risk customers (31.78%) facing erosion of lifetime value. (2) Lack of Predictive Intelligence—without customer health scoring, the organization cannot proactively intervene before customers leave. (3) Segmentation Gap—customer segments exist but lack actionable segment-specific churn and retention strategies. (4) Visibility Gap—executive teams lack real-time, comprehensive dashboards showing customer health and revenue-at-risk metrics.

BUSINESS OBJECTIVES

Key objectives include: identify customer churn drivers and at-risk segments with 95%+ accuracy; quantify revenue at-risk from churn (target: ₹300M+); develop product performance insights by category and geography; create interactive executive dashboards for real-time monitoring; provide actionable retention strategies with measurable ROI.

DATASET OVERVIEW

Table	Records	Key Fields	Grain
Fact_Orders	29,984	Order_ID, Customer_ID, Product_ID, Sales_Amount, Profit	One row per transaction
Dim_Customers	5,000	Customer_ID, Name, Age, City, Segment, Registration_Date	One row per customer
Dim_Products	300	Product_ID, Category, Sub_Category, Unit_Price	One row per SKU
Dim_Supportive_Cal	5,000	Customer_Status, CLV, Age_Group, Repeat_Flag	Calculated features per customer

Key Data Characteristics: Date Range: January 2, 2022 – December 31, 2025 (4-year window). Geographic Coverage: 32 Indian states and union territories. Product Categories: 8 major categories spanning 300 SKUs. Customer Demographics: Ages 18-65, balanced gender distribution. Payment Methods: 5 different payment types.

DATA CLEANING & PREPARATION

Data Quality Assessment

Issue	Records Affected	Resolution
Missing Customer Segment	32 (0.64%)	Imputed from CLV quartiles
Negative Profit Orders	58 (0.19%)	Retained as valid; flagged for review
Inconsistent Date Formats	Minimal	Standardized to ISO 8601
Duplicate Orders	0	All Order_IDs unique

Orphaned IDs	0	Full referential integrity verified

Data Transformation Steps: (1) Date Standardization—converted all date fields to ISO 8601 format. (2) Currency Formatting—verified all monetary values in Indian Rupees with 2 decimal precision. (3) ID Validation—ensured consistent naming conventions. (4) Missing Value Handling—confirmed no null values in key metrics. (5) Duplicate Detection—verified Order_ID uniqueness. (6) Outlier Analysis—identified extreme values as valid outliers requiring retention.

DATA MODELING (POWER PIVOT RELATIONSHIPS)

Database Schema Architecture

The data model follows a classical star schema approach with 1 fact table and 3 dimension tables. Relationships enable multi-dimensional analysis across customer, product, and time dimensions. Cross-filter directions are optimized for dashboard responsiveness.

Table Type	Table Name	Primary Key	Record Count
Fact	Fact_Orders	Order_ID	29,984
Dimension	Dim_Customers	Customer_ID	5,000
Dimension	Dim_Products	Product_ID	300
Dimension	Dim_Supportive_Cal	Customer_ID	5,000

Relationship Specifications: (1) Fact_Orders ↔ Dim_Customers: One-to-Many on Customer_ID (1 customer has many orders). (2) Fact_Orders ↔ Dim_Products: One-to-Many on Product_ID (1 product appears in many orders). (3) Dim_Customers ↔ Dim_Supportive_Cal: One-to-One on Customer_ID (calculated features per customer). Benefits include complex multi-dimensional analysis, avoidance of data redundancy, and instant dashboard responsiveness.

FEATURE ENGINEERING

Advanced analytics requires transforming raw data into meaningful features for churn prediction and segmentation.

1. CUSTOMER STATUS (Churn Indicator)

Status	Definition	Count	% of Base
Active	Purchase within 90 days	2,809	56.18%
At Risk	90-180 days since last purchase	1,589	31.78%
Churned	180+ days inactive	602	12.04%

2. CUSTOMER LIFETIME VALUE (CLV)

Formula: Sum of Profit from all orders by customer. Distribution: Average CLV ₹187,963; Median ₹88,690; Range ₹113 to ₹2,550,671. Top 20% generate 62% of lifetime profit. Business Impact: CLV guides marketing budget allocation and retention prioritization.

3. AGE GROUP SEGMENTATION

Age Group	Range	Count	% of Base
Young	18-30	1,895	37.9%

Adult	31-44	1,048	21.0%
Senior	45-65	1,554	31.1%
Older	Age 65+	503	10.1%

Insight: Young customers represent 38% of base; Senior 31%. Middle-aged (Adult) has lowest representation (21%), indicating untapped segment opportunity.

4. REPEAT CUSTOMER FLAG

Definition: Customers with Order_Count ≥ 2. Metrics: Repeat Customers: 4,194 (83.88%); One-Time Buyers: 806 (16.12%); Average Orders: 6.0; Max Orders: 60. Insight: 83.88% repeat rate indicates exceptionally strong customer satisfaction.

KEY PERFORMANCE INDICATORS (KPIs)

KPI	Definition	Actual Value	Target
Revenue	Sum of Sales_Amount	₹939.8M	₹1.0B
Profit	Sum of Profit from all orders	₹253.5M	₹300M
Orders	Count of Order_ID	29,984	35,000
Customers	Count of unique Customer_ID	5,000	6,000
AOV	Revenue / Orders	₹31,344	₹40,000
CLV	Sum of Profit per customer	₹187,963 avg	₹250,000
Churn Rate	(Churned / Total Customers) %	12.04%	<9%
Retention Rate	(Active + At-Risk / Total) %	87.96%	>91%
Profit Margin	Profit / Revenue	26.96%	>28%

DASHBOARD ANALYSIS

Dashboard 1: EXECUTIVE DASHBOARD

Key Metrics Displayed: 6 KPIs (Revenue, Profit, Orders, Customers, Churn, Retention) plus monthly trend analysis. Monthly Revenue Trend reveals strong seasonality (Dec peak: ₹250.1M, Jan trough: ₹18.7M). Geographic Analysis shows top 5 cities drive 23% of revenue, indicating concentration risk. Order Status shows 85% delivery rate with 10% returns and 5% cancellations.

Dashboard 2: SALES & CUSTOMER PERFORMANCE DASHBOARD

Focus: Product & Customer Value Metrics. Top 10 Products by Revenue (MacBook Pro: ₹26.1M) vs. Profit shows margin variation. Sales by Category (Electronics: 46%, Books: 28%) highlights product mix concentration. Top 10 Customers by CLV reveals 80/20 rule—top 200 customers (4%) likely generate 50% of profit. Order Status Trend (Dec growth in both sales and profit) shows consistent profitability despite order volume fluctuations.

Dashboard 3: CHURN & RETENTION DASHBOARD

Focus: Customer Health Monitoring. Active vs At-Risk vs Churned breakdown visualizes segmentation. Churn by Segment (Bronze: 11.6%, Silver: 9.3%, Gold: 8.4%, Platinum: 12.6%) shows that higher-tier segments also churn, pointing to engagement issues beyond price. Monthly Active vs Churned Customers trend shows churned count increasing from Jan (73) to peak (122 in July), then stabilizing. Revenue by Status shows Active customers (79%) generate majority revenue, emphasizing retention ROI.

KEY INSIGHTS & FINDINGS

1. REVENUE & PROFITABILITY: Annual revenue ₹939.8M with 27% net margin. Monthly seasonality peaks in December (₹250.1M) and troughs in January (₹18.7M). Strong profitability demonstrates pricing power and operational efficiency.

2. GEOGRAPHIC CONCENTRATION: Top 5 cities generate ₹216.8M (23% of revenue); top 10 account for 41.5%. Concentration creates risk; expansion into tier-3 markets recommended.

3. PRODUCT CATEGORY DISPARITY: Electronics (46% revenue) dominates; Books (28%) strong second. Combined, top 2 categories drive 74% of revenue. Over-reliance creates margin vulnerability.

4. CUSTOMER SEGMENT VALUE: Platinum (4.4%) and Gold (19%) show strong CLV; Bronze (43%) dominates headcount but lower individual value. Implement tiered service models accordingly.

5. CHURN CRISIS: 1,589 at-risk customers (31.78%) show purchase decline. If churned, CLV loss exceeds ₹300M. Immediate action required: segment-specific retention campaigns.

6. FULFILLMENT QUALITY: 85% delivery rate; 10% return rate above industry average (7-9%). Quality improvement reduces returns and improves profitability.

7. PRODUCT PERFORMANCE: Top 10 products (of 300) generate 19.1% of revenue. Low performers (200 products = 10% revenue) are inventory inefficient. SKU rationalization opportunity.

BUSINESS RECOMMENDATIONS

Recommendation 1: AGGRESSIVE CHURN MITIGATION

Actions: Launch immediate re-engagement campaign for 1,589 at-risk customers; offer segment-specific incentives (15% discount Bronze, 10% Silver, VIP for Platinum); implement weekly nurture sequences; establish customer health scoring model. Expected Impact: Recover 20% of at-risk customers (318) → ₹60M+ CLV retained; reduce churn rate to 9.5%.

Recommendation 2: PRODUCT PORTFOLIO OPTIMIZATION

Actions: Consolidate low-performing SKUs; increase inventory for top 20 products; launch high-margin category growth (Books 23.8%, Fashion 54%); develop bundling strategy. Expected Impact: Margin improvement 100-150 bps; AOV growth 8-12%.

Recommendation 3: GEOGRAPHIC EXPANSION

Actions: Target tier-2 cities with low penetration; partner with local logistics; customize product mix by geography. Expected Impact: Revenue growth 15-20% YoY; diversify city concentration to 25%.

Recommendation 4: SEGMENT DIFFERENTIATION

Actions: Implement tiered loyalty programs by segment (Platinum: concierge, Gold: premium support, Silver: rewards, Bronze: cost-efficient). Expected Impact: Improve retention 5-15% by segment; increase CLV 20%.

Recommendation 5: QUALITY & LOGISTICS IMPROVEMENT

Actions: Reduce return rate from 10% to 7%; reduce cancellations from 5% to 3%; establish 95%+ on-time delivery SLA. Expected Impact: ₹15M+ profit recovery; improve NPS by 8-10 points.

CHALLENGES & DATA QUALITY ISSUES

Data Quality Challenges: Inconsistent date formats resolved through standardization. 32 missing customer segments (0.64%) imputed from CLV quartiles. Negative profit orders (0.19%) retained as valid business transactions. Profit margin outliers (-122% max) flagged for vendor review but retained for analysis.

Technical Limitations: Excel Power Pivot can handle ~1M rows (current 30K well within limits). Dashboard reflects point-in-time analysis; for live operational dashboards, integrate with transaction database. Current analysis is descriptive; next step is predictive churn modeling using logistic regression or random forest.

SKILLS DEMONSTRATED

Skill Category	Skills Demonstrated	Proficiency
Data Modeling	Star schema design, Power Pivot relationships, dimensional modeling	Advanced
Data Cleaning	Data validation, duplicate detection, missing value imputation, outlier analysis	Advanced
Feature Engineering	RFM analysis, customer segmentation, CLV calculation, churn prediction features	Advanced
Excel Functions	SUMIF, COUNTIF, INDEX/MATCH, complex formulas, pivot tables	Expert
Visualization	Dashboard design, KPI cards, trend analysis, geospatial analysis, drill-down interactivity	Advanced
Analytics	Customer segmentation, churn analysis, product performance, geographic analysis	Advanced
Business Acumen	Strategic recommendations, business impact quantification, ROI analysis	Advanced
Communication	Executive reporting, data storytelling, technical documentation	Professional

TOOLS & TECHNOLOGIES USED

Tool/Technology	Usage	Level
Excel	Data storage, pivot tables, formulas, dashboards	Expert
Power Query	Data transformation, ETL, cleaning	Advanced
Power Pivot	Dimensional modeling, relationships, DAX calculations	Advanced
Pivot Tables	Multi-dimensional analysis, aggregation	Expert

Pivot Charts	Data visualization, trend analysis	Advanced
Slicers	Interactive dashboard filtering, drill-down	Advanced
Conditional Formatting	Visual highlighting, data quality checks	Intermediate
VBA/Macros	Automation (if used)	Intermediate

CONCLUSION & NEXT STEPS

Summary: This project successfully transformed raw e-commerce data into actionable business intelligence. Built star-schema data model integrating 4 tables; engineered 5 advanced features; created 3 interactive dashboards; identified critical business issues; delivered 5 strategic recommendations with quantified business impact.

Key Findings: (1) Highly profitable (₹939.8M revenue, 27% margins) but churn crisis (12% rate). (2) 1,589 at-risk customers represent ₹300M+ CLV at risk. (3) Electronics dominates (46%) but margin compression evident. (4) Strong repeat customer ratio (83.88%) demonstrates product-market fit. (5) Geographic and product concentration create business risk.

Next Steps: IMMEDIATE (0-30 days): Launch churn recovery campaign; establish weekly monitoring. SHORT-TERM (1-3 months): Build predictive churn model; implement loyalty programs; begin SKU rationalization. MEDIUM-TERM (3-12 months): Scale geographic expansion; integrate live data feeds; deploy machine learning models.

With disciplined execution of these recommendations, SwiftMart can improve retention by 3-5%, grow revenue by 15-20%, and expand margins by 100-150 bps within 12 months.